

Circuit Simulation Lab:6

Decoders

Introduction

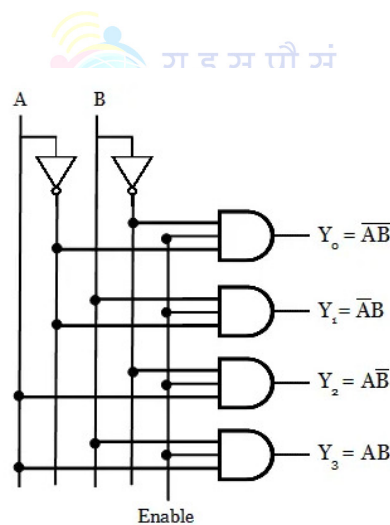
The purpose of this experiment is to introduce the design of simple combinational circuits, in this case Decoders. A decoder is a multiple-input, multiple-output combinational logic circuit. It converts the n bit data inputs into the coded 2^n outputs. It decodes the information hidden by the encoder.

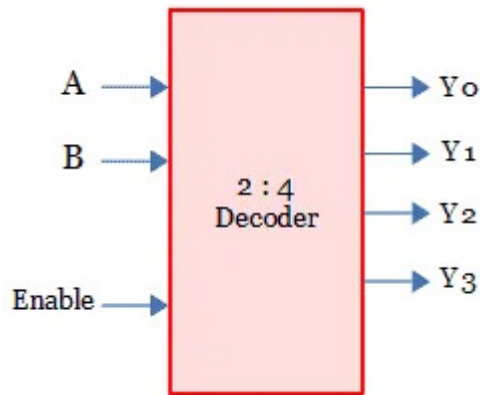
Software tools Requirement

Modelsim (Siemens)

Xilinx Vivado

Logic Diagram





2:4 Decoder

Inputs			Outputs			
EN	A	B	Y ₃	Y ₂	Y ₁	Y ₀
0	x	x	0	0	0	0
1	0	0	0	0	0	1
1	0	1	0	0	1	0
1	1	0	0	1	0	0
1	1	1	1	0	0	0

Truth Table

Describe the above in Verilog HDL and capture the Waveforms

1. Dataflow modeling
2. Behavior modeling
3. Structural modeling (Design a 4 to 16 decoder by using two 3 to 8 decoders)
4. Design a 4 to 16 decoder using case statements in Verilog